



The HyperXite team with their winning entry in the Hyperloop Pod Competition (Source: <https://arstechnica.com>)

the pressure is equalised with the station, before being released into the transit area. The doors of the capsule would open so that passengers could disembark. The luggage pod would be quickly unloaded by the Hyperloop staff or separated from the capsule so that baggage retrieval does not interfere with the capsule turnaround.

Once vacated, the capsule will be rotated on a turntable, and aligned for re-entry into the Hyperloop tube. The departing passengers, and their pre-loaded luggage pod, would then enter the capsule. The capsule would be moved forward into the exit airlock, where the pressure is lowered to the operating level of the Hyperloop, and then sent on its way.

The role of 'IT' in shaping the Hyperloop

Way back in September 2013, Ansys Corporation conducted computational fluid dynamics simulations to model the aerodynamics of the capsule and shear stress forces that the capsule would need to

endure. Simulation revealed that the capsule shape required considerable fine-tuning to prevent supersonic airflow. It was also inferred that the gap between the tube wall and capsule would need to be increased.

Later in October 2013, the development team of the OpenMDAO software framework (which serves as the backbone for subsystem integration, optimisation and workflow management) released a partially completed conceptual open source model of parts of the Hyperloop's

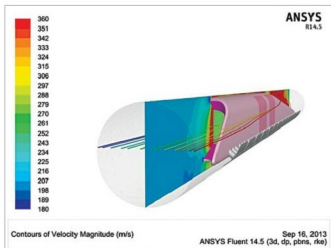
propulsion system. However, there were some hindrances and the model required tweaking.

In November 2013, MathWorks analysed the tweaking inputs and concluded that there was a high probability for that route to be possible. The analysis focused on the acceleration felt by passengers and the essential deviations from public roads in order to keep the accelerations within limits.

In January 2015, a paper based on the NASA OpenMDAO open source model emphasised again the need for a bigger-diameter tube and a lower cruise speed closer to Mach 0.85. It suggested doing away with the on-board heat exchangers based on thermal models of the interactions between the compressor cycle, tube and ambient environment. The fully integrated system model was constructed using OpenMDAO—a Python-based framework for design and optimisation of highly coupled systems. This work focused on the aerodynamics and thermodynamics of the passenger pod itself. OpenMDAO gave the means to manage growing model complexity in an efficient and flexible way.

The presented models were built exclusively using freely-available, open source tools. As the Hyperloop design evolved, more and more design parameters were required to be managed and optimised.

In 2015, an undergraduate Hyperloop team from the University of California (UCI) got together to compete in a chain of contests sponsored by SpaceX. UCI's team, called HyperXite, won a technical excellence award in a design competition held in 2016, and the team had to actually build the thing in time for the January 2017 pod contest at SpaceX's headquarters in Los Angeles.



ANSYS technology at work: A Hyperloop simulation using software in progress (Source: www.ansys-blog.com)